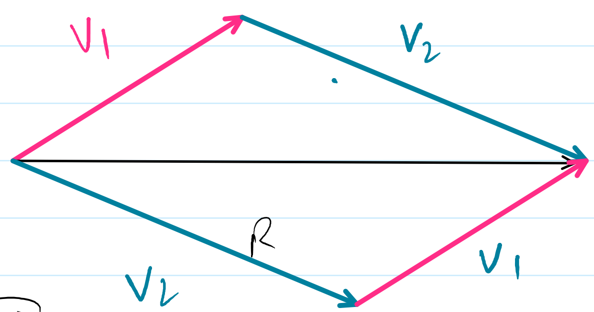
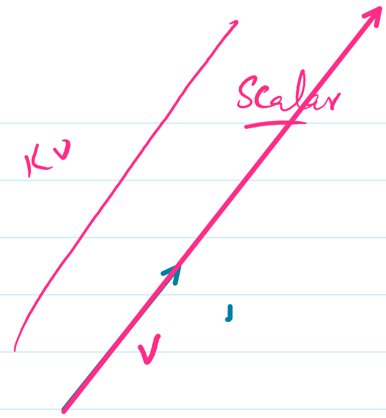
Lecture #32,33(Vector Spaces)

Friday, May 15, 2026 10:12 AM

Vector (magnitude + direction)

Scalar magnitude

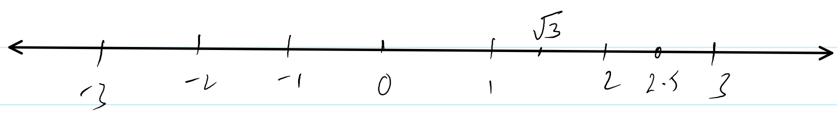




Real Vector Space

All Components are Real.

$R \in Q \cup \hat{Q} \Rightarrow$



$\frac{p}{q}$ SKAKAL mein Na hoon

Complex vector space

$\sqrt{-2}$ $\sqrt{-1} = i$

$a + \hat{i} b$

1) $\boxed{\pm 3, \pm 2, \pm 5 \; --}$ $\rightarrow$ Rational $\Rightarrow \frac{5}{1}$; Integers.

2) $\pm \frac{2}{3}, \pm \frac{5}{2}, \pm \frac{7}{2}$ "

3) 0.45, 2.359, ... $\frac{0.45}{100}$, $2.359 \Rightarrow \frac{2359}{1000}$ p/q

4) $0.424242 \ldots = 0.\dot{4}\dot{2}$ AJMCK

$x = 0.424242\ldots$ $\times 100$ $\frac{42}{99}$

$100x = 42.424242\ldots$

$x = 0.424242\ldots$

$99x = 42.$ $0.4242\ldots$ $x = \frac{42}{99}$

$x = 0.1252525\ldots$) $\times 10$

$x = 2354444\ldots$) $\times 1000$

$x = 0.1252525\cdots$) $\times 10$

$10x = 1.252525\cdots$) $\times 100$

$1000x = 125.252525\cdots$

$10x = 1.252525\cdots$

$990x = 124$

$x = \dfrac{124}{990}$

$x = 235444\cdots$) $\times 1000$

$1000x = 235.4444\cdots$ $\times 10$

$10{,}000x = 2354.444\cdots$

$1000x = 235.444\cdots$

$9000x = 2119$

$x = \dfrac{2119}{9000}$

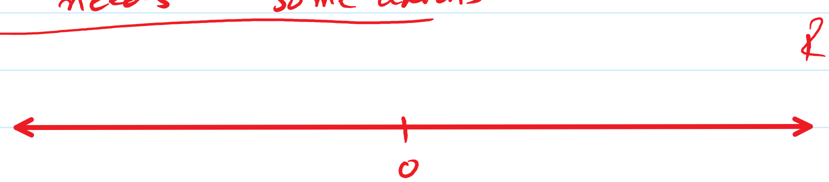
$$\begin{array}{r} 2354 \\ 235 \\ \hline 2119 \end{array}$$

$x = 0.5679879 8$

$= \dfrac{56742}{99900}$

$$\begin{array}{r} 56798 \\ 56 \\ \hline 56742 \end{array}$$

Vector space needs some axioms



2, 3	$2+3=5 \in \mathbb{R}$	Closed (closure)
3	$3\times 7 = 21 \in \mathbb{R}$	Scalar multiplication
$4+7$	$= 7+4 = 11$	Commutative
$4+(7+3)$	$= (4+7)+3 = 14$	Associative
$5+0$	$= 5$	additive Identity
$5+(-5)$	$= 0$	Additive Inverse

$(3+7)(5) = 3\times5 + 7\times5 = 50$ distributive property of addition over multip

$3(7+5) = 3(7)+3(5)$ Distributive property of multiplication over addition

4.1 Real Vector Spaces

Let V be a nonempty set of objects, on which two operations are defined:

4.1 Real Vector Spaces

Let V be a nonempty set of objects, on which two operations are defined:

a) Addition, b) Multiplication by scalars

With the following properties:

1. If $\vec{u}$ and $\vec{v}$ are elements in V, then $\vec{u}+\vec{v}$ is in V. (V is closed under addition)
2. $\vec{u}+\vec{v}=\vec{v}+\vec{u}$, for all $\vec{u}$, $\vec{v}$ in V. (holds Commutative Law)
3. $\vec{u}+(\vec{v}+\vec{w})=(\vec{u}+\vec{v})+\vec{w}$ (holds Associative Law)
4. There is an object $\vec{0}$ in V, called the zero vector for V such that $O+\vec{u}=\vec{u}+O=\vec{u}$, for each $\vec{u}$ in V. (have Additive Identity)
5. For each $\vec{u}$ in V, there is an object $-\vec{u}$ in V, called a negative of $\vec{u}$, such that $\vec{u}+(-\vec{u})=-\vec{u}+\vec{u}=O$. (have Additive Inverse)
6. If k is any scalar and $\vec{u}$ is any object in V, then $k\vec{u}$ is in V. (Closed under Scalar Multiplication).
7. $k(\vec{u}+\vec{v})=k\vec{u}+k\vec{v}$
8. $(k+m)\vec{u}=k\vec{u}+m\vec{u}$
9. $k(m\vec{u})=(km)\vec{u}$
10. $1\vec{u}=\vec{u}$ (have Multiplicative Identity)

then V is called a vector space and the objects in V *are* ***vectors***.

Example 1: Let $V=R^2=\{(x,y);\, x,y\in R\}$, prove that V is a vector space under the usual operations of addition and scalar multiplication defined by:

$$\vec{u}+\vec{v}=(u_1,u_2)+(v_1,v_2)=(u_1+v_1,\vec{u}_2+\vec{v}_2)$$

$$k\vec{u}=k(\vec{u}_1,\vec{u}_2)=(k\vec{u}_1,k\vec{u}_2)$$

Solution:

1. V is closed under addition. (as defined)
2. Let $\vec{u}=(\vec{u}_1,\vec{u}_2),\vec{v}=(\vec{v}_1,\vec{v}_2)$

$$\vec{u}+\vec{v}=(\vec{u}_1,\vec{u}_2)+(\vec{v}_1,\vec{v}_2)$$

$$=(\vec{u}_1+\vec{v}_1,\vec{u}_2+\vec{v}_2)$$

$$=(\vec{v}_1+\vec{u}_1,\vec{v}_2+\vec{u}_2)$$

$$=(\vec{v}_1,\vec{v}_2)+(\vec{u}_1,\vec{u}_2)=\vec{v}+\vec{u}$$

3. Let $\vec{u}=(\vec{u}_1,\vec{u}_2),\vec{v}=(\vec{v}_1,\vec{v}_2),\vec{w}=(\vec{w}_1,\vec{w}_2)$ $\quad (u_1, u_2)+((v_1+w_1, v_2+w_2))$

$$= (\vec{v}_1, \vec{v}_2) + (\vec{u}_1, \vec{u}_2) = \vec{v} + \vec{u}$$

3. Let $\vec{u} = (\vec{u}_1, \vec{u}_2), \vec{v} = (\vec{v}_1, \vec{v}_2), \vec{w} = (\vec{w}_1, \vec{w}_2)$

$$(\vec{u} + \vec{v}) + \vec{w} = (\vec{u}_1 + \vec{v}_1, \vec{u}_2 + \vec{v}_2) + (\vec{w}_1, \vec{w}_2)$$
$$= (\vec{u}_1 + \vec{v}_1 + \vec{w}_1, \vec{u}_2 + \vec{v}_2 + \vec{w}_2)$$
$$= (\vec{u}_1 + (\vec{v}_1 + \vec{w}_1), \vec{u}_2 + (\vec{v}_2 + \vec{w}_2))$$
$$= (\vec{u}_1, \vec{u}_2) + (\vec{v}_1 + \vec{w}_1, \vec{v}_2 + \vec{w}_2)$$
$$= \vec{u} + (\vec{v} + \vec{w})$$

$(u, u_2) + [(v_1 + w_1, v_2 + w_2)]$

$u + [(v_1, v_2) + (w_1, w_2)]$

$u + (v + w)$

4. Let $\vec{u} = (\vec{u}_1, \vec{u}_2), \vec{0} = (0,0)$

$$\vec{u} + \vec{0} = (\vec{u}_1, \vec{u}_2) + (0,0) = (\vec{u}_1, \vec{u}_2) = \vec{u}$$

5. Let $\vec{u} = (\vec{u}_1, \vec{u}_2)$, then there exist $-\vec{u} = (-\vec{u}_1, -\vec{u}_2)$,

$$\vec{u} + (-\vec{u}) = (\vec{u}_1 + (-\vec{u}_1), \vec{u}_2 + (-\vec{u}_2)) = (\vec{u}_1 - \vec{u}_1, \vec{u}_2 - \vec{u}_2)$$
$$= (0,0) = \vec{0}$$

$k(\vec{u} + \vec{v})$

6. V is closed under scalar multiplication. (as defined).

7. $k(\vec{u} + \vec{v}) = k\left((\vec{u}_1, \vec{u}_2) + (\vec{v}_1, \vec{v}_2)\right)$

$$= k(\vec{u}_1 + \vec{v}_1, \vec{u}_2 + \vec{v}_2)$$
$$= (k\vec{u}_1 + k\vec{v}_1, k\vec{u}_2 + k\vec{v}_2)$$
$$= (k\vec{u}_1, k\vec{u}_2) + (k\vec{v}_1, k\vec{v}_2)$$
$$= k(\vec{u}_1, \vec{u}_2) + k(\vec{v}_1, \vec{v}_2)$$
$$= k\vec{u} + k\vec{v}$$

$K((u_1, u_2) + (v_1, v_2))$

$K[(u_1 + v_1), (u_2 + v_2)]$

$= (Ku_1 + Kv_1, Ku_2, Kv_2)$

$= (Ku_1, Ku_2) + (Kv_1, Kv_2)$

$= K(u_1, u_2) + K(v_1, v_2)$

$= Ku + Kv$

8. $(k + m)\vec{u} = (k + m)(\vec{u}_1, \vec{u}_2)$

$$= (k\vec{u}_1 + m\vec{u}_1, k\vec{u}_2 + m\vec{u}_2)$$
$$= (k\vec{u}_1, k\vec{u}_2) + (m\vec{u}_1, m\vec{u}_2)$$
$$= k(\vec{u}_1, \vec{u}_2) + m(\vec{u}_1, \vec{u}_2)$$
$$= k\vec{u} + m\vec{u}$$

9. $k(m\vec{u}) = k\left(m(\vec{u}_1, \vec{u}_2)\right) = (km\vec{u}_1, km\vec{u}_2)$

$$= km(\vec{u}_1, \vec{u}_2) = km(\vec{u})$$

10. $1\vec{u} = 1(\vec{u}_1, \vec{u}_2) = (\vec{u}_1, \vec{u}_2) = \vec{u}$

As the set V satisfies all the properties, so V is vector space.

Example 3: Let $V = R^2$, under the usual operations of addition defined by:

$$\vec{u} + \vec{v} = (\vec{u}_1, \vec{u}_2) + (\vec{v}_1, \vec{v}_2)$$
$$= (\vec{u}_1 + \vec{v}_1, \vec{u}_2 + \vec{v}_2)$$

$K\vec{U} = K(U_1, U_2)$

$= (KU_1, 0)$

And if k is any scalar number, then define

$k\vec{u} = k(\vec{u}_1, \vec{u}_2) = (k\vec{u}_1, 0)$ ✓

Not a vector space

The addition operation is standard one on R^2, but the scalar multiplication is not.

Check whether V is vector space or not?

Solution: All properties of addition are satisfied. (Check it by yourself)

Let's check the properties of scalar multiplication.

$K(\vec{u} + \vec{v}) = KU + KV$

1. Let $\vec{u} = (u_1, u_2)$ in V, then $k\vec{u} = k(u_1, u_2) = (ku_1, 0) \in V$.
2. Let $\vec{u} = (u_1, u_2), \vec{v} = (v_1, v_2)$

$$k(\vec{u} + \vec{v}) = k\left((u_1, u_2) + (v_1, v_2)\right)$$
$$= k(u_1 + v_1, u_2 + v_2)$$
$$= (ku_1 + kv_1, 0)$$
$$= (k\vec{u}_1, 0) + (k\vec{v}_1, 0)$$
$$= k(\vec{u}_1, 0) + k(\vec{v}_1, 0)$$
$$\neq k\vec{u} + k\vec{v}$$

$U = (u_1, u_2)$ $\vec{v} = (v_1, v_2)$

$K(\vec{U} + V) = K[(U_1, U_2) + (v_1, v_2)]$

$= K[(U_1 + V_1, U_2 + V_2)]$

$= (K(U_1 + V_1), 0)$

$= (KU_1 + KV_1, 0 + 0)$

$= (KU_1, 0) + (KV_1, 0)$

$= K(U_1, 0) + K(V, 0)$

$\neq KU + KV$

As the 7th property does not satisfied So it's not a vector space.

Example 4:

Check whether V is vector space or not?

V = The set of all pairs of real numbers of the form $(x, 0)$. i.e. $\{(x, 0); x \in R\}$

with the standard operations on R^2.

$$\vec{u} + \vec{v} = (\vec{u}_1, \vec{u}_2) + (\vec{v}_1, \vec{v}_2) = (\vec{u}_1 + \vec{v}_1, \vec{u}_2 + \vec{v}_2)$$

$$k\vec{u} = k(\vec{u}_1, \vec{u}_2) = (k\vec{u}_1, k\vec{u}_2)$$

1. $U = (U_1, 0)$ $V = (V_1, 0)$

$U + V = (U_1, 0) + (V_1, 0)$

$= U_1 + V_1, 0 + 0$

$= (U_1 + V_1, 0)$ closed

(2) $K(u_1, 0) = (Ku_1, 0)$ Scalar Multiplicatio

(3) $\vec{u} = (u_1, 0)$, $\vec{v} = (v_1, 0)$

$\vec{u} + \vec{v} = (u_1 + v_1, 0+0)$

$= (v_1 + u_1, 0+0)$

$= (v_1, 0) + (u_1, 0)$

$\vec{u} + \vec{v} = \vec{v} + \vec{u}$

(iii) $\vec{u} = (u_1, 0)$ $\quad \vec{0} = (0,0) \Rightarrow$ Yes

$\vec{u} = (u_1, 0)$ $\quad -\vec{u} = (-u_1, 0)$ $\quad (\vec{u} + u_1, 0)$

$(0, 0)$

$(K+m)\vec{u}$ ———————— $K\vec{u} + m\vec{u}$

$= (K+m)(u_1, 0)$

$= ((K+m)u_1, 0)$

$(Ku_1, 0) + (mu_1, 0)$

$= K(u_1, 0) + m(u_1, 0)$

$= Ku + mu$

$K(u_1, 0) + m(u_1, 0)$

$(Ku_1, 0) + (mu_1, 0)$

Example 5: Check whether V is a vector space or not.

V = set of all pairs of real numbers of the form (x, y), where $x \geq 0$, i.e.

$V = \{(x, y); x \geq 0, y \in R\}$

With standard operations on R^2.

$(u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2)$

$K(u_1, u_2) = (Ku_1, Ku_2)$

* additive Identity

additive Inverse

$u = (u_1, u_2)$ $\quad -u = (-u_1, -u_2)$

No additive inverse

Not a vector space

Example 6: Show that the set of all pairs of real numbers of the form $(x, 1)$ with the operations:

$$(x, 1) + (x', 1) = (x + x', 1)$$

& $k(x, 1) = \left(k^2 x, 1\right)$ is not a vector space.

① $u = (u_1, 1)$, $v = (v_1, 1)$

$u + v = (u_1 + v_1, 1)$

$k(u_1, 1) = (k^2 u_1, 1)$

$u + v = (u_1, 1) + (v_1, 1) = (u_1 + v_1, 1)$

$= (v_1 + u_1, 1)$

$= (v_1, 1) + (u_1, 1)$

$v + u$

$(0, 0)$ $(x, 1)$

does not Exist

NOT A Vector Space